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DIGITAL ASSIGNMENT - 1

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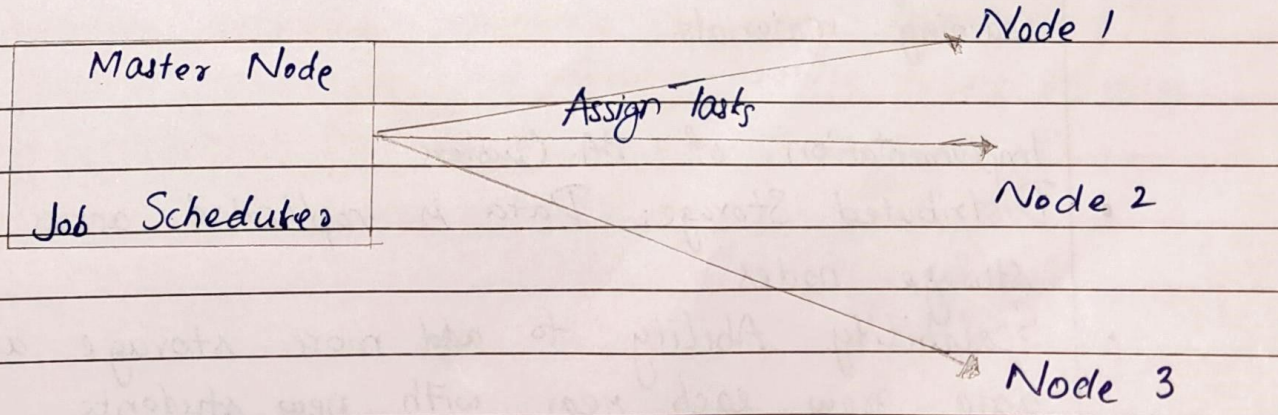
Qn.1 VIT University can ~~ob~~enhance its academic, research and admistrative operations by implementing different types of clusters computing system. Each of the type plays a vital role in managing large-scale student data, supporting computationally intensive research projects, and ensuring uninterrupted access to online resources.

High Performance Computing (HPC) Clusters)

High Performance clusters are essential for research that involves complex computations, such as simulations, deep learning, scientific ~~knowledge~~ modelling. These clusters consists of multiple nodes working in parallel to deliver high computational power.

Implementation of HPC Clusters

- Hardware: Multi-node with high-speed interconnects (eg. InfiniBand, Gigabit Ethernet)
- Software: Use of parallel computing frameworks like MPI and Open MP
- Resource Allocation: Job Scheduling through Slurm or PBS (Portable Batch System)



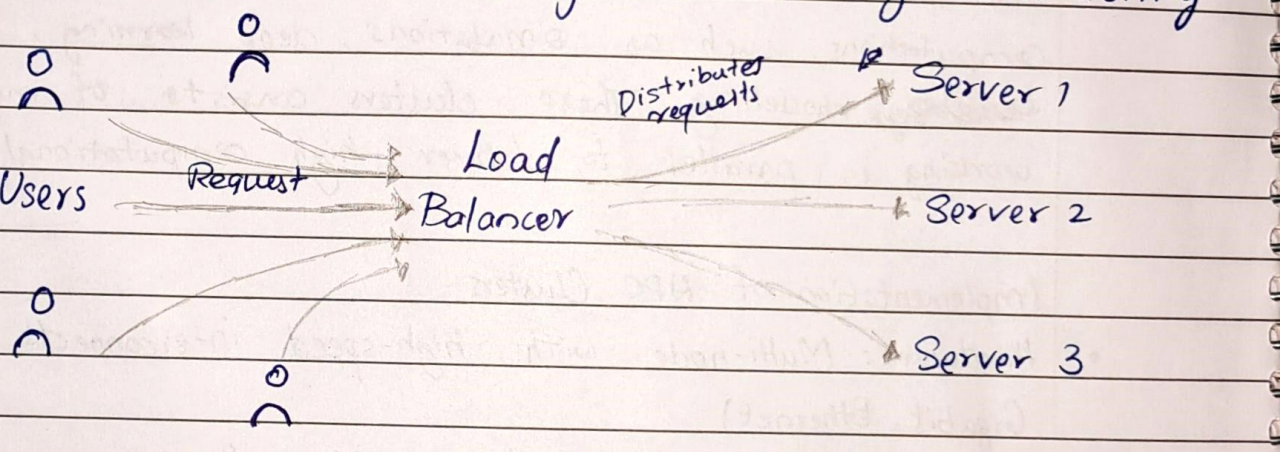
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Load-Balancing Clusters

A Load-balancing Clustor ensures efficient distribution of workloads among servers, preventing overload and improving response time for academic portals and research tools.

Implementation of Load-Balancing Clusters

- Load Balancer: A central controllers distributes incoming requests to multiple backend servers.
- Server Pool: A collection of web and data base servers handling university applications
- Fault Tolerance: Redundancy ensures high availability



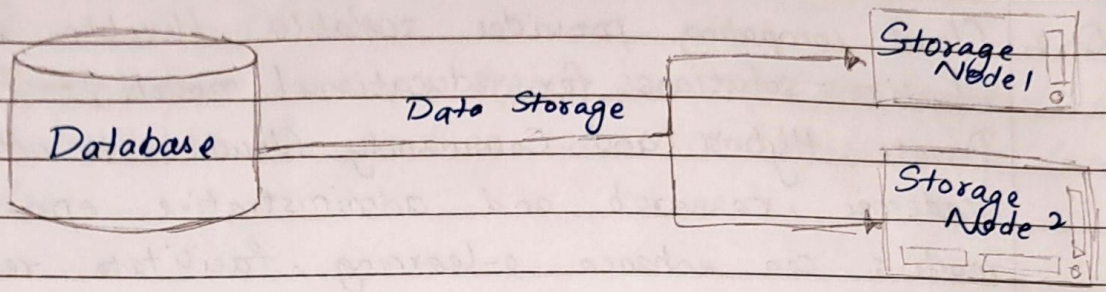
High-Availability Clusters

High availability clusters ensures secure, redundant and high-speed data management. It is vital for handling student records, research data and online learning materials.

Implementation of HA Clusters

- Distributed Storage: Data is replicated across multiple storage nodes.
- Scalability: Ability to add more storage as the data grow each year with new students
- Data Integrity: Redundancy and automated backups.

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Benefits of Cluster Computing for VIT

- Enhanced Research Capabilities: HPC Clusters enable complex computations for various research domains.
- Reliable Online Services: Load-balancing clusters ensure smooth academic and administrative operations.
- Efficient Data Management: HA clusters provide a secure and scalable solution for handling large amounts of student and research data.

By implementing HPC, Load-balancing and HA clusters, VIT University can significantly improve efficiency, reliability and performance across academic, research and administrative activities. These systems ensure a seamless computing experience Computing experience for students, researchers, and faculty while future-proofing the institutions technological infrastructure.

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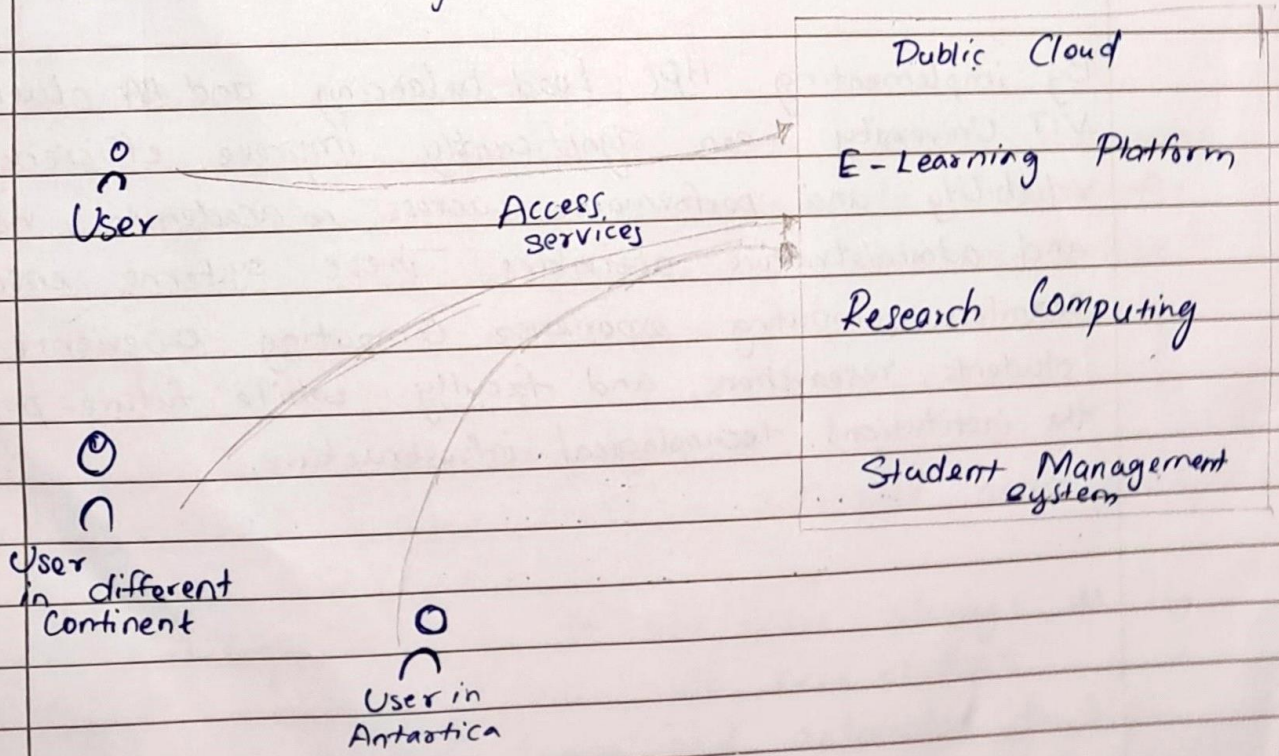
Qn.2. Cloud computing provides scalable, flexible and cost-effective solutions for educational models, — Public, Private, Hybrid and Community Clouds — to optimize academic, research and administrative operations. These models can enhance e-learning, facilitate research collaborations, ensure data security, and support digital transformations.

Public Cloud

Public cloud services are managed by third-party providers offering scalability and cost efficiency.

Implementation of Public Cloud for VIT

- E-Learning platforms: Hosting platforms like Moodle, Coursera, custom LMS on AWS, Google Cloud or Azure.
- Research Computing: Access to cloud-based HPC for simulations and AI/ML workloads.
- Administrative Portals: Cloud-based student management and examination system.



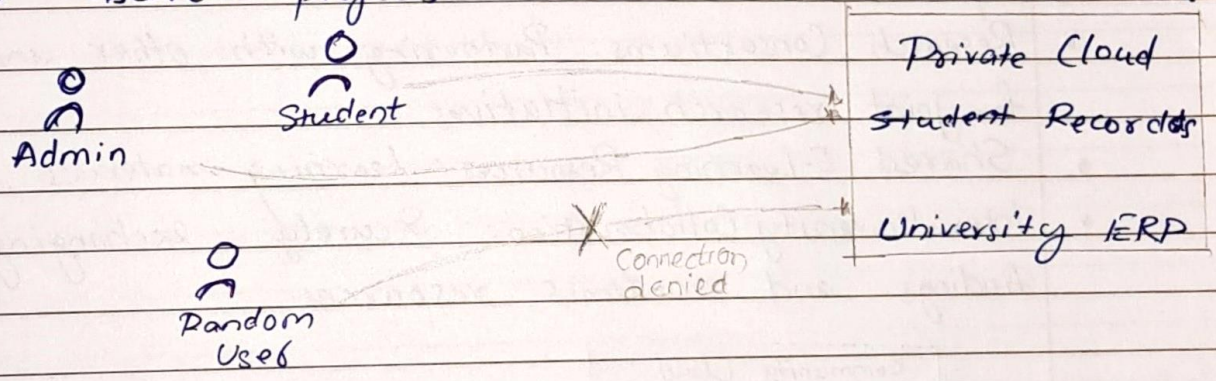
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Private Cloud

Private cloud ensures higher security and control, making it suitable for handling sensitive student and faculty data

Implementation

- Data security: Hosting student records, financial data, and faculty research.
- Custom internal Applications: Hosting ERP, HR and internal collaboration tools.
- High - Performance Computing : Dedicated private servers for research projects



Hybrid Cloud

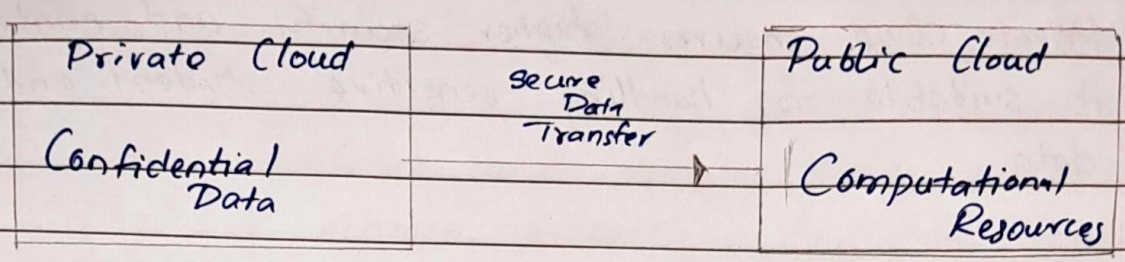
Hybrid cloud combines public and private cloud advantages, providing flexibility while ensuring security.

Implementation

- Disaster Recovery: Backing up critical systems to the public cloud while running primary operations on a private cloud.
- Research Collaboration: Storing confidential data on a private cloud while utilizing public cloud for computational tasks.
- Scalable Infrastructure: Expanding computing resources dynamic -ally based on academic needs.

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Hybrid Cloud

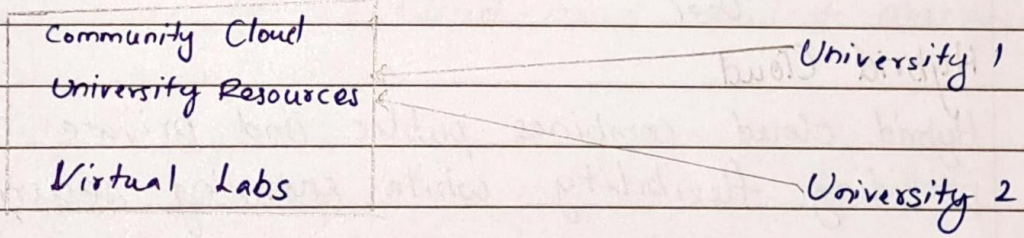


Community Cloud

A community cloud serves multiple institutions with shared interests, providing cost-effective collaboration.

Implementation

- Research Consortiums: Partnering with other universities for joint research initiatives.
- Shared E-Learning Resources: Learning materials, virtual labs etc.
- Inter-University Collaboration: Securely exchanging research findings and academic resoarces.



Benefits

- Scalability: Elastic computing power to accomodate growing academic and research demands
- Cost-effectiveness: Reduced IT infrastructure costs through cloud based solutions.
- Security & Compliance: Ensuring data protection through private and hybrid cloud models.